

Synergy Wallet

Technical Whitepaper & System Specification

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Canonical Design Principles

The following three principles are non-negotiable design constraints that supersede all earlier drafts. Every section of this specification is written in compliance with these principles.

A — SXCP North Star (Bridgeless)

- SXCP never executes transactions; it only coordinates execution that happens elsewhere.
- Source-chain execution occurs natively; destination-chain execution occurs natively.
- SXCP is bridgeless: it refuses to create any intermediate custody domain (no pooled vaults, no mint/release wrappers, no synthetic representations).
- Threshold Attestation Mode attests to observed chain facts (finalization, inclusion, scope). It is not delegated execution.

B — Seed-Theft Resistance (Non-Negotiable)

- Possession of a seed phrase (mnemonic) alone **MUST** be insufficient to spend or to complete wallet recovery.
- The mnemonic is treated as one factor in a multi-factor authority construction (Factor-Bound Root Authority, §7.2.3).
- The second factor (F2) must be high entropy, non-memorable, and not stored with the mnemonic.
- All execution keys are derived from $WRA = HKDF(MS \parallel F2)$, not from MS alone.

C — PQC Alignment (Aegis Authority Boundary)

- Aegis governs all authoritative protocol roles: identity, routing governance, attestations, recovery receipts, and audit artifacts.
- The platform baseline for authoritative signing artifacts is ML-DSA (FIPS 204). FN-DSA (FIPS 206) is permitted only where explicitly specified by role or policy and accompanied by a clear interoperability rationale.
- External-chain execution remains canonical to each chain until that chain supports PQ verification. The wallet explicitly distinguishes Aegis-enforced properties from audited-only properties.
- No subsystem may silently substitute non-Aegis signature primitives for Aegis-governed roles.

0. Preface and Document Scope

0.1 Purpose and Audience

This document is the canonical internal technical specification for the Synergy Wallet system. It supersedes and fully integrates all prior drafts, including V1 and V2 addenda. It is written to serve simultaneously as:

- an engineering reference for implementation and review,
- an architectural description for security and protocol analysis,
- a disclosure substrate for provisional and non-provisional patent filings.

It is not intended as marketing material or a public-facing whitepaper.

0.2 Disclosure Status and IP Considerations

This document is unredacted and contains system-level details that materially contribute to the novelty of the Synergy Wallet architecture. Public distribution prior to formal IP filing is not permitted.

0.3 Terminology and Conventions

- "Identity" refers to Synergy-layer authority, not external-chain accounts.
- "Execution" refers exclusively to native blockchain transaction signing.
- "Routing" refers to address resolution and intent direction, not asset movement.
- "Wallet" refers to the composite system, not a single keypair.
- "WRA" denotes the Factor-Bound Root Authority secret from which all operational keys are derived.
- "F2" denotes the high-entropy second factor required alongside the mnemonic-derived master seed.
- "Aegis" denotes the Synergy post-quantum cryptographic governance subsystem.

1. Abstract

Synergy Wallet is a self-custodial, identity-centric, multi-chain wallet system designed to eliminate irreversible asset loss, collapse fragmented blockchain user experiences, and enable protocol-level cross-chain execution without custodial bridges, delegated signing, or wrapped assets.

Unlike conventional wallets that bind identity, authority, and execution to a single static private key, Synergy Wallet introduces a layered control architecture that strictly separates identity authority, routing intelligence, and execution capability into independently governed but cryptographically linked planes. This separation enables survivable custody, deterministic multi-chain interoperability, and cryptographic longevity including full compatibility with post-quantum security models.

A defining security property distinguishes this architecture from all prior work: possession of the BIP-39 mnemonic phrase alone is computationally insufficient to spend funds or complete wallet recovery. Execution authority requires both the master seed and an independent high-entropy second factor, bound together through the Factor-Bound Root Authority (WRA) construction. This property is enforced cryptographically, not by convention.

This document specifies the internal architecture, cryptographic construction, state models, operational flows, and recovery semantics of Synergy Wallet in sufficient detail to support both engineering implementation and intellectual-property protection.

2. Problem Statement and Design Motivation

2.1 Structural Failures of Existing Wallet Architectures

Most existing cryptocurrency wallets suffer from architectural flaws rather than implementation bugs. These flaws include:

Single-Point Catastrophic Failure.

Loss or compromise of a seed phrase results in permanent loss of identity, authority, and assets. There is no structural distinction between 'attacker who took a photograph of your seed' and 'authorized owner.'

Identity–Execution Conflation.

A single private key simultaneously represents user identity, governance authority, and transaction execution power. These roles are collapsed into one cryptographic secret.

Chain Fragmentation.

Users must manage incompatible address formats, signing curves, derivation standards, and mental models across every supported chain.

Unsafe Interoperability.

Cross-chain transfers rely on custodial bridges, wrapped assets, and opaque trust assumptions that introduce systematic custody risk.

Cryptographic Obsolescence.

Most wallets cannot evolve toward post-quantum security without breaking compatibility or custody guarantees.

2.2 Why UX Abstractions Are Insufficient

Many modern wallets attempt to address usability and safety concerns by introducing user-interface abstractions layered atop legacy cryptographic assumptions. Examples include hiding seed phrases after initial setup, aggregating balances across chains in a single dashboard, abstracting gas fees, and relabeling addresses with contacts or ENS-like systems.

While these measures may improve short-term usability, they do not address the underlying structural failures of wallet architecture. A cosmetic solution modifies how a user interacts with a system. A structural solution modifies what the system is allowed to do.

In conventional wallets, Identity = Private Key, Authority = Private Key, and Execution = Private Key. No amount of UI abstraction can change the fact that loss or compromise of that key permanently destroys identity, authority, and execution capability simultaneously. Synergy Wallet rejects the premise that these roles must be unified.

2.3 Design Goals and Explicit Non-Goals

2.3.1 Primary Design Goals

Synergy Wallet is designed to satisfy the following goals simultaneously:

1. **Survivable Self-Custody.** The system must remain usable and secure even after partial cryptographic failure (e.g., loss of execution keys, device loss, or mnemonic exposure).

2. **Seed-Theft Resistance.** Possession of the mnemonic alone must not confer spend capability. This is enforced at the cryptographic derivation layer, not the UI layer.
3. **Native Chain Correctness.** All external transactions must be indistinguishable from transactions produced by chain-native wallets.
4. **Identity-Centric Interaction.** Users should interact using persistent identities rather than ephemeral addresses.
5. **Deterministic Multi-Chain Support.** All supported chains must be derivable from a single root authority without custodial infrastructure.
6. **Cryptographic Longevity.** The system must support staged migration toward post-quantum cryptography without breaking compatibility.

2.3.2 Explicit Non-Goals

Synergy Wallet explicitly does not aim to:

- act as a custodian or quasi-custodian,
- delegate signing authority to servers or relayers,
- replace external chain execution semantics,
- abstract away security-critical user consent,
- enforce protocol-level control over user assets,
- recover assets tied to execution keys that were irretrievably lost.

These exclusions are intentional and enforced architecturally.

3. Architectural Overview

3.1 The Three-Plane Model

Synergy Wallet is structured around three strictly separated planes, each responsible for a distinct class of system behavior. This separation is structural, not optional.

Plane	Responsibility	What It Cannot Do
Identity	Define authority and ownership	Execute transactions or sign chain-native TXs
Routing	Resolve intent and destination (UMA)	Authorize execution or hold private keys
Execution	Move value on external chains natively	Govern identity, modify routing, or delegate authority

3.2 Explicit Architectural Invariants

The system enforces the following invariants at all times:

7. No single key can define identity and execute value transfer.
8. Identity authority cannot sign external-chain transactions.
9. Routing data cannot initiate or authorize execution.
10. Execution keys cannot modify identity or routing state.
11. Loss of any single cryptographic secret cannot permanently destroy future asset control.
12. Possession of a mnemonic phrase alone cannot authorize spending (WRA requirement).

3.3 Authority Separation and Trust Boundaries

Each plane operates under a different trust model:

- Identity Plane: Trusted for governance and intent, but intentionally powerless over execution.
- Routing Plane: Public, inspectable, revocable, and intentionally non-authoritative.
- Execution Plane: Fully authoritative for value movement, but blind to identity semantics.

No plane can escalate privileges into another.

3.4 Comparison to Conventional Wallet Models

Dimension	Conventional Wallet	Synergy Wallet
Identity	= Private Key	Independent plane (SIK); cannot execute
Authority	= Private Key	Governed by Identity Plane; separated from execution
Execution	= Private Key	Derived from WRA; disposable instruments
Recovery	Restore the key	Restore authority via identity anchoring; no key resurrection
Cross-Chain	Per-chain silos / bridges	Protocol coordination via SXCP (bridgeless)
Custody	Self (fragile) or bridge	Explicitly none; all execution is chain-native
PQC	Future replacement	Parallel authority today via Aegis/ML-DSA
Seed Exposure	= Total loss	Insufficient alone; F2 required (WRA)

4. Identity Plane — Authority Without Execution

4.1 Purpose and Scope

The Identity Plane defines who controls authority within the Synergy Wallet system, independent of how or where value is executed. It is the root of governance, but it is intentionally powerless over asset movement.

Core Responsibility.

The Identity Plane defines wallet ownership, governs routing configuration (UMA), controls recovery and rotation mechanisms, authorizes protocol-level coordination (e.g., SXCP participation), and anchors human-readable identity (SNS).

Explicit Prohibition.

The Identity Plane cannot sign external-chain transactions, derive or access execution private keys, initiate value movement, bypass local execution policies, or retroactively assert control over past execution authority. This prohibition is enforced structurally, not by convention.

4.2 Synergy Identity Key (SIK)

The Synergy Identity Key (SIK) is the primary authority key for a Synergy Wallet identity. It represents governance authority, not execution authority.

Cryptographic Scheme.

Under V2 alignment with Aegis, the SIK baseline scheme is ML-DSA (FIPS 204). FN-DSA (FIPS 206) may be used where explicitly specified by role policy. Classical schemes (ECDSA, EdDSA) remain supported for compatibility contexts. The choice of scheme does not affect the Execution Plane, ensuring cryptographic agility.

Authority Scope.

The SIK has authority over SNS name ownership, UMA routing records, recovery configuration, policy configuration, and protocol attestations. It explicitly does not have authority over external-chain signing, transaction finalization, or asset custody.

4.3 Identity State Model

A Synergy Identity exists in one of several explicit, finite states. Identity state is cryptographically enforced and auditable.

State	Description
Active	Normal operation; full authority in effect
Restricted	Limited operations due to policy violation, suspected compromise, or recovery guard
Recovery-Pending	Recovery initiated; time-lock in effect; UMA updates may be frozen
Recovery-Confirmed	Quorum or time-lock satisfied; awaiting generation increment
Recovered	New execution generation activated; identity returns to Active
Decommissioned	Identity permanently retired; no further state transitions possible

State transitions are monotonic within defined flows, may require time delays or quorum approval, cannot be bypassed by execution keys, and cannot be retroactively applied.

4.4 Recovery Authority and Guardianship

Recovery in Synergy Wallet does not mean reconstructing lost private keys. Recovery means restoring identity authority sufficient to govern future routing and execution derivation. During recovery, existing execution keys are not recovered, past transactions remain immutable, future execution authority is re-derived deterministically from a new epoch, and routing records are rotated forward-only.

4.5 Synergy Address and SNS

A Synergy Address is a Bech32m-encoded identifier derived from identity material. It is non-executable and non-signing. It serves as a pointer, not a destination. SNS (Synergy Naming System) binds human-readable names (e.g., `alice.syn`) to Synergy Addresses; it resolves only to Synergy Addresses, never to execution endpoints, IP addresses, or contracts.

4.6 Identity Plane Invariants

- Identity $\neq$ Execution
- Authority $\neq$ Signing Power
- Recovery $\neq$ Key Reconstruction
- Names $\neq$ Addresses
- Governance $\neq$ Custody

5. Routing Plane — Intent Without Authority

5.1 Purpose and Scope

The Routing Plane determines where value should be delivered, without possessing the ability to move that value. It is the system's intent resolution layer, not a control layer. It maps Synergy identities to chain-specific public addresses, enables identity-centric payments across heterogeneous chains, supports address rotation without asset migration, and enables recovery without execution key reconstruction.

The Routing Plane cannot sign transactions, initiate execution, hold private keys, bypass local execution policy, or assert authority over identity or execution planes. Routing is advisory, never authoritative.

5.2 Universal Meta-Address (UMA)

A Universal Meta-Address (UMA) is a cryptographically governed routing construct that maps a Synergy Identity to one or more chain-native public addresses. Conceptually: (Synergy Identity, ChainID, Version) → Public Address. UMA represents where value should be sent, not who can move it.

UMA records are explicitly versioned. Each update increments a monotonic version counter, invalidates prior routing entries, may be subject to time-locks or quorum rules, and is auditable and irreversible. UMA evolution is forward-only; previous routing paths are never overwritten silently.

A canonical UMA record contains the Synergy Identity reference, Chain identifier (ChainID), public destination address, version number, validity window (optional), revocation status, authorization signature from the Identity Plane, and update metadata. UMA records are public by design. No sensitive material is stored in UMA.

5.3 Authority Model and Governance

UMA updates are authorized exclusively by the Identity Plane. Execution keys cannot modify UMA records. Routing records cannot modify identity authority. UMA updates may require single-signature identity approval, multi-signature quorum, guardian consent, or time-delayed activation.

5.4 Routing Plane Invariants

- Routing ≠ Authority
- Resolution ≠ Execution
- Public Data ≠ Custody
- Rotation ≠ Asset Migration
- Recovery ≠ Key Reconstruction

6. Execution Plane — Authority Without Identity

6.1 Purpose and Scope

The Execution Plane is responsible for moving value on external blockchains according to each chain's native rules. It is the only plane capable of signing transactions. At the same time, it is intentionally blind to identity semantics, recovery governance, and routing intent beyond what is locally resolved.

The Execution Plane cannot modify identity authority, update UMA routing records, resolve SNS names independently, override local policy, or delegate signing authority off-device. Execution is powerful but narrow by design.

6.2 Execution Keys as Disposable Authorities

In conventional wallets, execution keys are treated as permanent identity anchors. In Synergy Wallet, execution keys are treated as replaceable, deterministic instruments authorized by identity. This is a foundational departure. Execution keys are not identity. They are tools derived from the Factor-Bound Root Authority (WRA).

6.3 Deterministic Chain-Native Key Derivation

For each supported blockchain, Synergy Wallet derives keys using that chain's canonical derivation scheme, without modification. All derivation begins from WRA, not from the master seed directly.

Chain Family	Curve / Scheme	Standard
Bitcoin & UTXO chains	secp256k1	BIP-32 / BIP-44
Ethereum & EVM chains	secp256k1	BIP-44
Solana	ed25519	SLIP-0010
Substrate-based	sr25519	Native substrate
PQC domains	ML-DSA / FN-DSA	Aegis (FIPS 204 / FIPS 206)

Execution keys are derived from execution-specific HKDF domains within WRA. Compromise of one chain's private key does not imply compromise of other chains, identity keys, routing authority, or recovery authority.

6.4 Execution Authorization Pipeline

Execution authorization follows a strict, ordered pipeline. If any step fails, execution aborts before signing.

13. User Intent Expression — the user explicitly initiates an action.
14. Routing Resolution — destination address is resolved via UMA.
15. Policy Evaluation — Local Policy Engine evaluates all constraints (hard gate).

16. Execution Key Selection — correct chain-specific key selected from WRA-derived domain.
17. Transaction Construction — native transaction assembled per chain rules.
18. Signature Generation — transaction is signed locally.
19. Broadcast — transaction submitted directly to the chain RPC.

6.5 Native Execution Guarantee

From the perspective of external chains, every transaction appears as: Sender Address → Recipient Address. There are no proxy contracts, no wrapped assets, no Synergy-controlled intermediaries, and no synthetic representations. Transactions are indistinguishable from those produced by chain-native wallets.

6.6 Execution Plane Invariants

- Execution ≠ Identity
- Signing ≠ Governance
- Key Loss ≠ Identity Loss
- Native Semantics ≠ Protocol Control
- Replaceable Keys ≠ Custodial Risk

7. Root Secret and Cryptographic Isolation

This section defines the wallet's cryptographic foundation, including the Factor-Bound Root Authority (WRA) construction that enforces seed-theft resistance as a non-negotiable architectural property.

7.1 Purpose of the Root Secret Architecture

The Root Secret architecture exists to solve a problem that most wallets silently ignore: determinism without isolation is dangerous, and isolation without determinism is unusable. Synergy Wallet is explicitly designed to achieve both.

A second, more critical property is also addressed here: in all prior wallet architectures, possession of the mnemonic phrase is equivalent to possession of the wallet. Synergy Wallet breaks this equivalence by requiring two independent factors for execution authority derivation.

7.2 Master Seed Construction

7.2.1 Mnemonic Creation

At wallet initialization, Synergy Wallet generates a standard BIP-39 mnemonic:

- Length: 12 or 24 words
- Entropy source: cryptographically secure RNG
- Optional user passphrase (BIP-39 extension)

The mnemonic itself is never used directly for key derivation beyond initial seed expansion.

7.2.2 Master Seed Derivation

The mnemonic is converted into a master seed (MS) using PBKDF2-HMAC-SHA512 with the standard BIP-39 iteration count and optional passphrase salt. The resulting master seed is a high-entropy binary secret treated as immutable. No keys are derived directly from MS alone.

7.2.3 Factor-Bound Root Authority (WRA) — Seed-Theft Resistance

This subsection specifies the wallet's core seed-theft resistance mechanism. Possession of the BIP-39 mnemonic alone **MUST NOT** confer spend-capable execution authority. If an attacker obtains the mnemonic phrase — by photograph, phishing, shoulder surfing, or backup compromise — that must not be sufficient to drain funds.

The V3 (unified) construction treats the mnemonic-derived master seed as an entropy component, not a master key. Execution authority is derived only from the Factor-Bound Root Authority secret (WRA), constructed from two independent factors under explicit policy.

Definition:

Let MS be the BIP-39 master seed derived via PBKDF2-HMAC-SHA512 as specified above.

Let F2 be a high-entropy second factor that is NOT human-memorable and is not stored alongside the mnemonic.

Compute: $WRA = HKDF(MS \parallel F2, \text{info}="synergy.wallet.wra.v1")$

All downstream cryptographic domains (Identity, Routing, Execution, PQC) are expanded from WRA using the HKDF domain separation rules defined in §7.3. MS alone is insufficient to derive any operational signing keys.

Acceptable F2 sources:

- Device enclave secret (Secure Enclave / TrustZone bound to a registered device)
- WebAuthn / passkey resident secret material (hardware-backed, non-exportable)
- Hardware security token seed (FIDO2 authenticator, hardware wallet)
- Guardian-supplied threshold share (recoverable without Synergy custody)

Required properties of F2:

- High entropy (≥ 128 bits effective)
- Non-exportable where possible (secure enclave / hardware-backed keystore)
- Must be recoverable without Synergy custody, via threshold mechanisms (guardians, secondary devices, or institutional co-signers)
- Must not be stored alongside or derivable from the mnemonic

Deployment Tiers:

Tier	Name	Description
1	Compatibility	Standard single-key EOA derivation (no seed-theft resistance). Wallet MUST label this tier clearly as not meeting the non-fatal seed compromise objective. Use only for legacy chain compatibility.
2	Factor-Bound Derivation	Execution keys derived from $WRA = HKDF(MS \parallel F2)$. Seed alone cannot spend; spending requires access to the second factor. This is the default deployment tier.
3	Threshold Execution	Threshold signing or chain-native scripts/accounts so that no single derived key can unilaterally spend. Provides defense-in-depth beyond Factor-Bound Derivation.
4	Chain-Enforced Accounts	Native scripts (UTXO) or contract accounts (EVM / account abstraction) enforce multi-factor or threshold authorization on-chain. Strongest tier; chain-dependent.

7.3 Domain Separation via HKDF

Synergy Wallet treats domain separation as a core security primitive, not an implementation detail. The WRA secret is expanded using HKDF into multiple cryptographically independent domains. Each domain is defined by a unique context string, a fixed domain identifier, and optional versioning and epoch data.

Canonical Domain Set:

- Identity Domain — used exclusively for Synergy Identity Keys and governance authority.
- Routing Domain — used for UMA control, routing attestations, and mapping governance.
- Execution Domains (per chain family) — separate domains for secp256k1, ed25519, sr25519, and future chain families.
- Post-Quantum Domains — separate domains for ML-DSA signatures, ML-KEM encapsulation, FN-DSA (policy-conditional), and threshold PQC attestations.

Each domain is non-invertible and non-derivable from any other domain.

7.4 Cryptographic Isolation Guarantees

Compromise of any one domain does not reveal WRA or MS, does not allow derivation of other domains, and does not escalate authority laterally. For example, a leaked Solana private key does not expose Bitcoin keys, Identity authority, UMA routing control, or Recovery keys. This property is structural, not assumed.

7.5 Temporal and Generational Derivation

Each domain supports generation counters, which allow new derivation epochs to be created without invalidating the root. Generation counters may increment during recovery events, device replacement, security incidents, or explicit rotation. Keys derived in generation n cannot be recomputed from generation $n+1$ data. This ensures lost keys stay lost, compromised keys remain isolated, and recovery does not resurrect insecure material.

7.6 Multi-Device Compatibility Without Shared Secrets

The domain architecture allows multiple devices to derive compatible keys without sharing private keys while remaining bound to a single identity. Devices derive keys independently using the same WRA material (established at initialization), the same domain context, and the same generation number. No device ever exports execution keys to another.

7.7 Root Secret Storage

WRA and MS are encrypted at rest, stored only locally, never transmitted, never logged, and never derived from SNS or UMA data. When accessed, WRA/MS is loaded only briefly for HKDF expansion and immediately zeroized afterward.

7.8 Root Secret Loss Scenarios

If WRA is lost but identity authority is preserved, recovery remains possible via identity-anchored regeneration (§10). F2 must be recoverable via the guardian/threshold mechanism declared at wallet setup. If both WRA and identity authority are lost, recovery is not possible. Synergy Wallet does not claim to eliminate all loss — it eliminates irreversible loss caused by single-factor failure.

7.9 Root Secret Invariants

- Determinism without key reuse
- Isolation without fragmentation
- Recoverability without resurrection
- Migration without re-custody
- Multi-chain compatibility without compromise
- Mnemonic alone is insufficient for execution authority

8. Transaction Models — SCETP

8.1 Purpose of Transaction Mode Separation

Synergy Wallet intentionally defines distinct transaction coordination modes rather than a single generalized transfer mechanism. Different transaction contexts impose fundamentally different requirements on trust assumptions, atomicity, latency, cryptographic coordination, and failure handling. Attempting to unify all transfers under a single abstraction is precisely what leads to custodial bridges, wrapped assets, opaque relayers, and silent trust escalation.

Two explicit and disjoint execution coordination protocols are defined: SCETP (same-chain) and SXCP (cross-chain). Each has strictly limited scope, explicit invariants, and non-overlapping responsibilities.

8.2 SCETP — Same-Chain External Transfer Protocol

SCETP is used exclusively when the source and destination chains are the same external blockchain (e.g., Solana → Solana, Bitcoin → Bitcoin). SCETP is not used for cross-chain settlement under any circumstances.

SCETP answers a single question: 'How do we let users send assets by identity without violating native chain semantics?' The answer is: resolve by identity, execute natively. Synergy participates in resolution and audit, never execution.

SCETP Execution Flow:

20. User Intent Declaration — user specifies asset type, amount, and recipient (SNS handle, Synergy address, or UMA reference).
21. Recipient Identity Resolution — SNS handle resolved to Synergy address, or direct Synergy address used.
22. Routing Resolution via UMA — Synergy identity resolved to chain-specific public address; deterministic and versioned.
23. Policy Evaluation (Local Only) — spending limits, velocity rules, MFA/guardian thresholds, asset-specific constraints. Any policy failure aborts execution before signing.
24. Execution Key Selection — chain-native private key derived deterministically from WRA domain; local to the device.
25. Native Transaction Construction — conforms exactly to chain rules; no wrappers or metadata requirements.
26. Local Signing & Broadcast — signature produced locally; transaction broadcast directly to chain RPC.
27. Optional Synergy Audit Receipt — Synergy records a reference event (sender/recipient identity, chain ID, external TX hash). Optional and non-authoritative.

SCETP Security Invariants:

- Synergy infrastructure never holds assets
- Synergy infrastructure never signs transactions
- UMA resolution never authorizes execution
- Identity compromise cannot move funds
- Routing compromise cannot bypass policy

9. SXCP — Synergy Cross-Chain Protocol

9.1 Scope and Core Design Principle

SXCP is used only when the source chain and destination chain differ. SXCP is not a transfer protocol in the traditional sense. It is a coordination protocol.

SXCP is built around one non-negotiable rule: SXCP never executes transactions. It coordinates execution that happens elsewhere. This single rule eliminates bridge custody risk, wrapped asset dependence, delegated signing, and silent trust escalation. SXCP operates entirely on commitments, proofs, and conditions.

SXCP is explicitly bridgeless: no intermediate custody domain is created or implied. There are no pooled vaults, no mint/release wrappers, no synthetic asset representations, and no wrapped tokens at any point in the coordination flow.

9.2 Cross-Chain Intent Model

SXCP treats a cross-chain transfer as a state machine, not a transaction. Each transfer progresses through explicitly defined states: Intent Declared → Source Commitment Observed → Conditions Verified → Destination Fulfillment Executed → Completion or Timeout Resolution. No state transition is implicit.

9.3 SXCP Operational Modes

9.3.1 Mode A — Atomic Swap Coordination (Trust-Minimized)

Mode A uses hash-locked commitments, asymmetric timeouts, and secret revelation as a fulfillment trigger. Properties: no relayer trust, fully on-chain verifiable, slower but maximally trust-minimized. Use cases: high-value transfers, adversarial environments, minimal trust assumptions.

9.3.2 Mode B — Threshold Attestation Coordination (High-Throughput)

Mode B uses witness registries, multi-party observation, and threshold signatures (PQC-capable). Properties: high throughput, low latency, Byzantine-fault tolerant. Use cases: retail transfers, high-volume routing, UX-optimized flows.

Critical distinction: relayers in Mode B attest to observed chain facts (finalization, inclusion, scope). They do not sign on behalf of users, do not delegate execution, do not hold assets, and do not execute destination-chain transactions. This is attestation of facts, not delegated execution.

9.4 Asset Flow Under SXCP

SXCP does not move assets itself. Source-chain execution occurs natively; destination-chain execution occurs natively; SXCP ensures coordination correctness between them. Assets remain on their native chains at all times. This preserves native asset semantics on all chains and eliminates custodial risk at the protocol level.

9.5 SXCP Failure Handling

- Timeout Reversion — assets return to original owner if fulfillment fails within the timeout window.
- Partial Observation Failure — transfer remains unresolved until quorum is reached; no silent loss.
- Relay Misbehavior — slashing, reputation loss, exclusion. Proceeds flow to compensation pool (§9.6).

No failure mode results in silent asset loss.

9.6 Compensation Pool and Remediation

SXCP includes a protocol-level remediation mechanism funded by protocol fee allocation and governed by the DAO. The compensation pool provides prospective (forward-looking) financial remediation for covered losses. It does not constitute retroactive state reversal or rollback, which SXCP never claims.

- Disbursement requires governance-verified evidence of covered loss.
- Coverage limits are published and auditable.
- Slashing proceeds from relay misbehavior flow to the compensation pool.
- Claims require cryptographic evidence of specific misbehavior via the Witness Registry.
- Emergency pause (scope-constrained, supermajority-gated, time-bounded) releases pending operations to source-chain owners; no cross-chain rollback is claimed.

9.7 Non-Custodial Guarantees

Property	SCETP	SXCP
Asset custody	Never	Never
Local signing	Always (WRA-derived keys)	Always (native to each chain)
Protocol coordination	No (direct native TX)	Yes (commitment/attestation only)
Wrapped assets	Never	Never
Intermediate custody domain	N/A	Never (bridgeless)
Delegated execution	Never	Never

10. Recovery Without Seed Reconstruction

10.1 Core Insight

Traditional wallets rely on a single assumption: possession of the seed phrase is equivalent to possession of the wallet. Synergy Wallet explicitly breaks this equivalence. The system is designed such that identity survivability, routing survivability, and future execution survivability do not depend on reconstructing historical execution keys. This is not a UX feature — it is a cryptographic and architectural property.

10.2 What Recovery Means in Synergy Wallet

Recovery in Synergy Wallet does not mean restoring old private keys, re-signing historical transactions, or reclaiming assets tied to irretrievably lost execution keys. Recovery means restoring identity authority, restoring routing control, and re-establishing future execution capability from a new WRA epoch.

10.3 Separation of Recoverable and Irrecoverable State

Irrecoverable State (by design):

- Execution private keys already used on external chains
- Assets already controlled by those keys
- Historical signing authority

This preserves blockchain immutability and prevents retroactive control claims.

Recoverable State (by architecture):

- Synergy identity authority (SIK)
- SNS name ownership
- UMA routing mappings
- Policy configuration
- Future execution derivation (new WRA epoch)

10.4 Recovery State Machine

State	Description
Active	Normal operation
Recovery-Pending	Recovery initiated; restrictions enforced; UMA updates may be frozen
Recovery-Confirmed	Quorum or time-lock satisfied; awaiting generation increment
Recovered	New execution generation activated from new WRA epoch
Restricted	Partial authority (e.g., routing frozen pending guardian approval)

Each transition is explicit, auditable, and irreversible.

10.5 Recovery Authorization Mechanisms

Recovery is identity-governed, not execution-governed. Recovery may require one or more of: identity signature, guardian quorum, multi-signature approval, time-delayed activation, external verification proofs. The exact requirements are configurable per identity at wallet setup.

10.6 Recovery Execution Flow

28. Recovery Initiation — identity enters Recovery-Pending state.
29. Restriction Enforcement — UMA updates frozen or limited; execution may be partially disabled.
30. Authorization Satisfaction — guardians / quorum / time-lock complete.
31. Generation Increment — derivation epoch increments; prior WRA material is deprecated.
32. New WRA Derivation — fresh F2 (re-established via threshold mechanism) combined with MS to produce new WRA.
33. New Execution Keys Derived — fresh chain-native keys derived from new WRA domains.
34. UMA Routing Rotation — routing updated to point to new addresses.
35. Recovery Finalization — identity returns to Active state.

No historical execution keys are reused. All authority epochs are forward-only.

10.7 Why This Is Not Custody or Escrow

Recovery does not imply Synergy holding keys, Synergy re-signing transactions, or Synergy reconstructing private keys. All recovery actions are authorization and derivation events, not execution. Execution authority remains local and device-bound. Synergy infrastructure is never in possession of user keys at any stage of recovery.

10.8 Recovery Invariants

- Identity $\neq$ Execution
- Recovery $\neq$ Key Resurrection
- Continuity $\neq$ Retroactivity
- Seed Loss $\neq$ Total Loss (F2 recovery via guardians provides path forward)
- Survivability $\neq$ Custody

11. Local Policy Engine

11.1 Purpose and Rationale

The Local Policy Engine exists to answer a single, critical question: Even if a valid WRA-derived execution key exists, should this transaction be allowed to proceed? In conventional wallets, policy is either nonexistent or advisory. In Synergy Wallet, policy is mandatory, pre-signature, and non-bypassable. The Policy Engine is the final authority before cryptographic execution, but it never signs transactions itself.

11.2 Policy as a Pre-Signature Hard Gate

All transactions — whether SCETP or SXCP-coordinated — must pass policy evaluation before any private key is used. If policy evaluation fails: no signature is produced, no transaction is constructed, no network broadcast occurs. There is no concept of 'sign now, review later.'

11.3 Policy Scope

The Policy Engine may enforce constraints across multiple dimensions simultaneously:

- Identity-scoped: maximum daily spend limits, total exposure caps per asset class, restrictions during recovery or restricted states, guardian approval thresholds.
- Chain-scoped: chain allow/deny lists, maximum gas or fee thresholds, time-of-day execution windows, network-specific risk profiles.
- Asset-scoped: per-transaction limits, velocity controls (rate limiting), whitelist/blacklist recipient logic, irreversible-asset protections.

11.4 Policy Governance

Policies are bound to identity state, not to devices or sessions. Policy rules are governed exclusively by the Identity Plane. Execution keys cannot modify policy. Routing data cannot modify policy. This prevents attackers from relaxing constraints even if execution keys are compromised.

Policy updates may require identity signature, multi-signature approval, guardian quorum, or delayed activation.

11.5 Policy Invariants

- Policy $\neq$ Advisory
- Policy $\neq$ Execution
- Identity governs policy
- Policy precedes signing
- No policy bypass paths

12. Post-Quantum Architecture (Aegis)

12.1 Core Design Principle

Post-quantum security cannot be confined to the network layer. If wallets remain classically brittle, the system fails at its most exposed edge. Synergy Wallet's PQC architecture is governed by Aegis and introduces post-quantum keys as parallel authority layers, not replacements. This enables PQC-secured identity and coordination, classical execution on legacy chains, and staged cryptographic migration — all without breaking backward compatibility.

The platform baseline for authoritative signing artifacts is ML-DSA (FIPS 204). FN-DSA (FIPS 206) is permitted only where explicitly specified by role or policy and accompanied by a clear interoperability rationale. No subsystem may silently substitute non-Aegis primitives for Aegis-governed roles.

12.2 Aegis Algorithm Suite and Wallet Roles

Six PQC algorithms are managed under Aegis. Four are active in Synergy Wallet; two are held as reserves due to operational size constraints.

Algorithm	Standard	Status	Role in Wallet
ML-DSA	FIPS 204	Active (baseline)	Identity authority baseline, SNS governance, UMA routing governance, SXCP attestations, protocol commitments, audit artifacts
ML-KEM	FIPS 203	Active	Encrypted wallet state, secure multi-device synchronization, guardian channels, recovery coordination, SXCP witness messaging
FN-DSA	FIPS 206	Active (conditional)	High-frequency validator attestations and wallet interactive sessions where lower latency is essential; only used where explicitly specified by policy; NOT used for high-stakes authoritative actions
HQC-KEM	NIST selected	Active (backup KEM)	Designated backup KEM activatable via governance if ML-KEM confidence is compromised; cryptographically independent of lattice assumptions (code-based)
SLH-DSA	FIPS 205	Reserve	Not active due to 8–50 KB signature sizes; operationally incompatible with consensus-frequency signing; maintained as reserve with hash-based security assumptions
CMCE-KEM	NIST alternate	Reserve	Classic McEliece; not active due to 261 KB–1 MB public keys; maintained as primary reserve KEM with independent hardness basis from both ML-KEM and HQC-KEM (oldest PQC proposal, 1978)

12.3 ML-DSA Key Roles

ML-DSA keys are derived from the Post-Quantum Identity domain under WRA and are used for:

- Synergy Identity Key (SIK) — baseline authoritative scheme
- SNS ownership assertions and updates
- UMA routing governance signatures
- SXCP witness attestations and threshold coordination
- Protocol-level commitments and recovery receipts
- Audit artifacts

ML-DSA keys never sign external-chain transactions. This preserves chain compatibility while strengthening internal authority.

12.4 ML-KEM Key Roles

ML-KEM keys are used for encrypted wallet state, secure multi-device synchronization, guardian communication channels, recovery coordination, and SXCP witness messaging. ML-KEM replaces classical encryption internally, not externally.

12.5 FN-DSA Key Roles and Constraints

FN-DSA (NTRU lattice-based, FIPS 206) is permitted only where explicitly designated by role or policy. It is suitable for high-frequency operations requiring lower latency (e.g., validator attestation sessions, interactive wallet sessions). It must not be used for high-stakes authoritative actions such as identity rotation, recovery initiation, or governance votes, where ML-DSA is required.

12.6 Hybrid Authority Model

Synergy Wallet supports hybrid verification where classical signatures validate execution authority and PQC signatures (ML-DSA baseline) validate identity and coordination authority. Both may coexist for the same logical operation. Example: a Solana transaction is signed classically; the accompanying intent and routing update are signed with ML-DSA; observers can verify both independently.

Hybrid verification is policy-governed and time-bounded, preventing indefinite complexity accumulation.

12.7 Forward Migration Strategy

36. Phase 1 — PQC for identity, routing, and coordination only. (Current state.)
37. Phase 2 — Hybrid verification (classical + PQC) for all operations.
38. Phase 3 — Optional PQC execution on chains that natively support it.

At no stage is backward compatibility broken.

12.8 PQC and Recovery Semantics

During recovery, ML-DSA identity keys may be used to reassert authority, routing updates require ML-DSA signatures, and execution key derivation remains classical for chain compatibility. This enables quantum-safe recovery even before external chains upgrade.

12.9 PQC Invariants

- PQC $\neq$ Execution Replacement
- Identity $\neq$ Execution Algorithm
- Migration $\neq$ Disruption
- Parallel Authority $\neq$ Custody
- Future Safety $\neq$ Retroactive Control
- ML-DSA is the authoritative baseline; FN-DSA is conditional and policy-gated

13. Threat Model and Security Analysis

13.1 Trust Domains

Domain	Trust Assumption
User device	Trusted while uncompromised at OS/hardware level
WRA / Master Seed	Trusted only under two-factor construction; mnemonic alone is untrusted
Execution keys	Trusted locally only; disposable instruments
Identity authority (SIK)	Semi-trusted; governed by policy; separated from execution
Routing data (UMA)	Public, untrusted for execution purposes; read-only advisory
Synergy infrastructure	Untrusted for custody; trusted only for protocol coordination
SXCP relayers	Byzantine-fault tolerant; trusted for attestation, not execution
External chains	Honest-majority consensus assumed per chain

13.2 Threats Explicitly Mitigated

Seed Phrase Theft.

Threat: attacker obtains the mnemonic phrase and drains funds. Mitigation: WRA construction requires F2 alongside MS; mnemonic alone cannot derive operational keys. Residual risk: assets tied to Tier 1 (compatibility) deployments without WRA are still vulnerable — these must be labeled accordingly.

Seed Phrase Loss.

Threat: user loses mnemonic and cannot recover. Mitigation: identity-anchored recovery via guardians allows re-derivation from a new WRA epoch. F2 must be independently recoverable via threshold mechanism.

Single-Key Compromise.

Threat: compromise of a single private key yields total control. Mitigation: three-plane separation with non-overlapping authority; WRA domain isolation prevents lateral compromise.

Custodial Bridge Failure.

Threat: bridge operators or smart contracts drain pooled assets. Mitigation: SXCP explicitly avoids asset custody, wrapping, and pooled liquidity. No intermediate custody domain exists.

Delegated Signing Abuse.

Threat: remote or delegated signers abuse authority. Mitigation: all execution signing is local, device-bound, WRA-derived, and policy-gated.

Routing Manipulation.

Threat: attacker alters routing to redirect funds. Mitigation: routing changes require ML-DSA identity authorization, are versioned, and remain non-authoritative for execution.

Quantum Adversaries (Future).

Threat: quantum computers break classical cryptography. Mitigation: parallel PQC authority (ML-DSA, ML-KEM) preserves identity, routing, and recovery even if execution keys degrade under quantum attack.

13.3 Threats Mitigated by Policy and Governance

- Rapid drain attacks — spending caps, velocity limits, time-based restrictions, guardian approvals.
- Social engineering — policy confirmation layers, recipient resolution transparency, time-delayed policy changes.
- Partial guardian compromise — quorum thresholds, diversity of guardians, time-lock activation.

13.4 Explicitly Out-of-Scope Threats

- Full device compromise at OS or hardware level (universal software wallet limitation).
- Physical coercion or legal compulsion.
- Consensus failure of external blockchains.

13.5 Failure Containment Model

Failure	Blast Radius
Mnemonic exposure	None (WRA requires F2; Tier 2+ unaffected)
Execution key loss	Single chain, single generation; identity/routing intact
F2 loss	Recovery path via guardian-threshold F2 re-establishment
Routing error	Execution aborts safely before signing
Identity compromise	Governance-gated recovery; execution keys separately derived
Relayer failure	Protocol timeout or reversion; compensation pool available

14. Patent-Relevant Technical Effects

This section enumerates technical effects that arise only from the specific architectural arrangements defined in this document. Each entry is written to support provisional patent filings, claim drafting, and examiner analysis against prior art.

14.1 Seed-Theft-Resistant Execution Authority (WRA Construction)

Technical effect: a cryptocurrency wallet where possession of the BIP-39 mnemonic phrase alone is computationally insufficient to derive any operational signing key. The novel construction $WRA = HKDF(MS \parallel F2)$ with $F2$ being high-entropy and non-collocated with the mnemonic introduces a two-factor derivation requirement enforced at the cryptographic root, prior to any domain expansion. No prior wallet architecture enforces this property without resorting to server-side custody or protocol-level key escrow.

14.2 Seed Phrase Regeneration Without Historical Key Resurrection

Technical effect: after loss of the mnemonic, a user may re-establish wallet control via identity-anchored recovery without reconstructing historical execution keys. New execution derivation proceeds from a new WRA epoch without resurrecting or re-enabling deprecated authority. This is technically distinct from seed backup, custodial recovery, and threshold key reconstruction — it is forward-only authority re-establishment.

14.3 Persistent Identity Independent of Execution Keys

Technical effect: a persistent cryptographic identity that survives loss of execution keys, device replacement, and cryptographic algorithm rotation without requiring custody or delegation. This breaks the long-standing wallet equivalence: private key = identity = asset control.

14.4 Three-Plane Separation with Cryptographic Non-Overlap

Technical effect: three non-overlapping cryptographic planes where no key material from one plane can derive keys for another, authorize actions in another, or modify authority in another. Failure in one plane is structurally contained. This is enforced by HKDF domain separation over WRA, not by convention.

14.5 Revocable, Identity-Governed Routing Fabric (UMA)

Technical effect: a versioned routing construct that maps cryptographic identity to chain-specific addresses, supports rotation without asset migration, and remains non-authoritative for execution. Neither hard-bound to identity nor custodially indirect. Public, revocable, and cryptographically governed.

14.6 Bridgeless Cross-Chain Coordination (SXCP)

Technical effect: cross-chain value coordination without any intermediate custody domain, wrapped asset, pooled vault, or delegated signing. Source-chain and destination-chain execution both occur natively. SXCP coordinates correctness between native executions without ever holding or moving assets itself.

14.7 PQC Parallel Authority Without Execution Disruption

Technical effect: post-quantum signing authority coexisting alongside classical execution semantics without requiring chain upgrades or wallet replacement. Identity, routing, and recovery are secured by ML-DSA; external-chain execution continues classically until chain-native PQC is supported.

14.8 Summary of Patent-Relevant Effects

- Seed-theft-resistant execution authority (WRA two-factor derivation)
- Forward-only seed regeneration without key resurrection
- Identity persistence independent of execution
- Recovery without historical key reconstruction
- Revocable routing without custody
- Cross-chain execution without wrapping or custody
- Deterministic multi-curve isolation
- PQC coexistence without disruption
- Survivable wallets under partial cryptographic failure

15. Conclusion — Canonical System Positioning

15.1 What Synergy Wallet Is — Precisely

Synergy Wallet is a cryptographically enforced separation of identity, routing, and execution that enables survivable self-custody, seed-theft resistance, deterministic multi-chain interoperability, and future-proof cryptographic migration without delegating signing authority or custody. Everything in this document follows from that definition.

15.2 Canonical Architectural Position

Dimension	Traditional Wallets	Smart-Contract Wallets	Synergy Wallet
Identity	Equals private key	Contract owner	Independent plane (SIK)
Execution	Same key as identity	Contract-mediated	Native, local, WRA-derived
Seed Exposure	= Total loss	Varies	Insufficient alone (WRA)
Recovery	Restore key	Contract logic	Identity authority; new WRA epoch
Cross-Chain	Per-chain silos / bridges	Bridges / wrappers	Protocol coordination (bridgeless)
Custody	Self (fragile)	Partial / implicit	Explicitly none
PQC	Future replacement	Not addressed	Parallel authority (ML-DSA)

15.3 Canonical Invariants

The following invariants define Synergy Wallet. If an implementation violates any of these, it is not a Synergy Wallet:

- Mnemonic alone does not authorize spending or recovery
- Identity authority never executes value
- Execution authority never governs identity
- Routing determines intent, not permission
- Recovery restores future control, not history
- Coordination never becomes custody
- Cryptographic migration never breaks compatibility
- ML-DSA is the authoritative signing baseline for protocol-governed roles

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